

## Scalability & Future Plan

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|----------------------|--------------|
| <b>Date</b>          | 12 July 2026 |
| <b>Team ID</b>       |              |
| <b>Project Name</b>  | Smart Lender |
| <b>Maximum Marks</b> | 1 Mark       |

### Current System Limitations:

| S.No | Limitation                                                                                                                            | Impact                                                                | Priority to Address<br>(High / Medium / Low) |
|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------|
| 1    | The application is designed primarily for local deployment and supports only a limited number of concurrent users.                    | Reduced performance when many users access the system simultaneously. | High                                         |
| 2    | The prediction model is based on Linear Regression, which may not capture complex relationships between health and lifestyle factors. | Prediction accuracy may decrease for diverse or large datasets.       | High                                         |
| 3    | The application currently stores limited user information and prediction history.                                                     | Difficult to perform long-term user analysis and trend monitoring.    | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                     | Proposed Upgrade / Solution                                                                                                        |
|------|--------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a limited number of users on a local Flask server.       | Deploy the application on cloud platforms such as Render, AWS, or Azure with load balancing to support a larger user base.         |
| 2    | Data Storage       | Uses CSV files and/or SQLite for storing data.                    | Migrate to MySQL or PostgreSQL to efficiently manage large datasets and user records.                                              |
| 3    | Performance        | Predictions are generated using a single Linear Regression model. | Optimize preprocessing, implement caching, and evaluate advanced machine learning models for faster and more accurate predictions. |
| 4    | Security           |                                                                   |                                                                                                                                    |

### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                                                                     | Target Timeline | Expected Impact                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------|
| Phase 1 | Improve the user interface and dashboard with advanced charts and responsive design.                                              | 1–2 Months      | Better user experience and easier interpretation of prediction results. |
| Phase 2 | Integrate advanced machine learning algorithms such as Random Forest, XGBoost, or Neural Networks to improve prediction accuracy. | 2–4 Months      | Higher prediction accuracy and better performance on diverse datasets.  |
| Phase 3 | Develop a mobile application and integrate cloud-based database storage with real-time synchronization.                           | 4–6 Months      | Increased accessibility and support for a larger number of users.       |

|         |                                                                                                                   |             |                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------|
| Phase 4 | Add AI-powered personalized health recommendations, wearable device integration, and real-time health monitoring. | 6–12 Months | Comprehensive well-being assessment, continuous monitoring, and personalized user guidance. |
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